

In the second project assignment, you will use the database design that you created in the first project assignment. You will design a queries that will meet the following requirements:

1. Create a query that will retrieve only selected columns from the selected table.
2. Create a query that will select user/person or similar table based on the email.
3. Create at least one UPDATE, INSERT, DELETE, and ALTER TABLE query.
4. Create a series of queries that will separately use the following:
NOTE: „;“ means perform both queries separately and „/“ means perform one or both queries
NOTE: If necessary, add additional columns or tables from project assignment 1. Perform the queries on meaningful data, e.g., do not perform AVG on id column.
 - (a) WHERE; WHERE with AND; WHERE with OR; WHERE with BETWEEN
 - (b) LIKE; NOT LIKE
 - (c) SUBSTRING; TRIM; CONCAT; COALESCE
 - (d) SUM; MIN; MAX; AVG;
 - (e) GROUP BY; GROUP BY and HAVING; GROUP BY, HAVING, and WHERE;
 - (f) UNION ALL / UNION; DISTINCT; COUNT; EXCEPT; INTERSECT
 - (g) LEFT JOIN; RIGHT JOIN; FULL OUTER JOIN; NATURAL JOIN
5. Use in one query: LEFT JOIN, GROUP BY, HAVING, ORDER BY, AVG and DISTINCT.
6. Create a query that will use any aggregate function and GROUP BY with HAVING.
7. Create a query that will join at least three tables and will use GROUP BY, COUNT, and HAVING.
8. Create a query that will return the data from an arbitrary table for the last one and half days (1day + 12 hours, i.e., 36 hours). Do not hard code the query (e.g., created_at > 7-11-2021)!
 - (a) Do it programmatically with DATE functions.
9. Create a query that will return data from the last month (starting from the first day of the month).
10. Write a select that will remove accents on a selected string (e.g., á will be converted to a).
 - (a) Beforehand, you will need to save data that contain accents in the database (e.g., save some Czech surname in the database that has accents).
11. Create a query for pagination in an application (use LIMIT and OFFSET).
12. Create a query that will use subquery in FROM.
13. Create a query that will use subquery in WHERE condition.
14. Create a query that will join at least five tables.
15. Modify the database from the first project assignment to improve integrity constraints (e.g., reduce the size for varchar columns or to add unique constraints, e.g., for email column).
 - (a) Set/improve cascading, explain places where you set/improve cascading and why?
16. Create database indexes (create it only on columns where it can make a sense).

- (a) Before you create a database index perform a query that will use WHERE condition on a column without index and then perform the same query on the column with index (note: use EXPLAIN keyword to note the differences – provide a comparison of the execution plans).
 - (b) Explain why it made sense to set indexes for these column(s).
 - (c) Explain the benefits and pitfalls of setting up the indexes.
17. Create arbitrary database procedure (consider some suitable case).
 - Briefly describe what your procedure is doing.
18. Create arbitrary database function (consider some suitable case).
 - Briefly describe what your function is doing.
19. Create arbitrary database trigger.
20. Create arbitrary database view (consider some complex case).
21. Create database materialized view (consider some complicated SQL query with several joins, aggregate function, GROUP BY with HAVING and complex WHERE condition). Explain why this materialized view is beneficial/needed.
22. If you use public schema create a new schema for your database called bds.
 - (a) Migrate all your data from public to bds schema.
 - (b) Revoke create on schema public from public.
 - (c) Explain why it might be useful to avoid public schema.
23. Create two roles my-app and my-script in your database.
 - (a) Assign for my-app privileges to SELECT, INSERT, UPDATE, and DELETE everything in arbitrary table in bds schema. Further, set for my-app the possibility to view only certain fields (e.g., without salary from „person“ or your „user“ object).
 - (b) For my-script assign a possibility to select only certain tables.
24. Encrypt selected database columns (using pgcrypto module).
 - Explain why you encrypted these columns.
 - Implement a SELECT query that decrypts and reads such encrypted data.
25. Describe the meaning of pg_hba.conf file in PostgreSQL. What can be configured here (focus on SSL and remote connections).
26. Describe what can be configured within postgresql.conf file in PostgreSQL (focus on deployment settings, SSL, logging, etc.).
 - Configure logging as follows: (logging_collector=on; Change log_filename so that new log file is generated each day). Consider additional suitable configurations for the logging (note the ones you considered).
 - Configure settings for the deployment of PostgreSQL (e.g., use the help of <https://pgtune.leopard.in.ua/>) for the DB version 15, OS Type: Linux, DB Type: Web application, RAM: 32 GB, Number of CPUs: 4, Number of Connections: 150, Data Storage: HDD storage.
 - Provide screenshots of your configurations and explain how you understand these configurations.

Important: Create a document that will note all these requirements together with the queries (or commands) performing them (please add also the attachments including the screen snippets of the query results). The document should be a single **.pdf** file with the following structure: Number of the query; description of the query; query or answer itself and the result (screen) of running it in DB administration tool.

This project assignment is performed by each student solely.

Create the GitLab project repository named „bds-project-assignment-2“ inside your student space (https://gitlab.com/but-courses/bpc-bds/student-projects/2022/<YOUR_SPACE>/bds-project-assignment-2) and upload the **.pdf** document. If you modify your database design from previous project assignment update the schema and upload the updated schema/scripts into this repository also.

This project assignment is for 6 points.

- **Target date: 14th of November (exclusive).**
- 3 for satisfying all the requirements (if any is not met, the number of points is set to 0).
- 0-3 based on the quality of a solution.